



Solving Radical Equations

Algebra Worksheet · Grade 8-10

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Learning Objectives

- Isolate the radical expression before squaring both sides
- Solve radical equations by squaring both sides to eliminate the square root
- Verify solutions and recognize when variables are inside vs. outside the radical

Solve each radical equation for the variable, showing all steps and isolating the variable.

1. Solve for x.

$$\sqrt{x} - 3 = 4$$

Answer: _____

2. Solve for x.

$$\sqrt{x} + 5 = 1$$

Answer: _____

3. Solve for x.

$$5 + \sqrt{x} = 7$$

Answer: _____

4. Solve for x.

$$\sqrt{x - 3} = 4$$

Answer: _____

5. Solve for m.

$$\sqrt{3m + 1} = \sqrt{2m + 6}$$

Answer: _____

6. Solve for x.

$$\sqrt{x + 7} = 5$$

Answer: _____

7. Solve for x.

$$\sqrt{2x + 1} = 3$$

Answer: _____



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8. Solve for x.

$$\sqrt{x} - 6 = 2$$

Answer: _____

9. Solve for x.

$$\sqrt{4x - 3} = 5$$

Answer: _____

10. Solve for x.

$$\sqrt{x - 1} + 2 = 6$$

Answer: _____



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Emphasize that students must isolate the radical before squaring, and remind them that squaring a negative number yields a positive result.

Solutions

1. Solve for x.

$$\sqrt{x} - 3 = 4$$

- Add 3 to both sides to isolate the square root
- The equation becomes square root of x equals 7
- Square both sides to eliminate the radical
- Seven squared is 49, so x equals 49

Answer: $x = 49$

2. Solve for x.

$$\sqrt{x} + 5 = 1$$

- Subtract 5 from both sides to isolate the square root
- The equation becomes square root of x equals negative 4
- Square both sides to eliminate the radical
- Negative 4 squared is 16, so x equals 16

Answer: $x = 16$

3. Solve for x.

$$5 + \sqrt{x} = 7$$

- Subtract 5 from both sides to isolate the radical
- The equation becomes square root of x equals 2
- Square both sides to eliminate the radical
- Two squared is 4, so x equals 4

Answer: $x = 4$

4. Solve for x.

$$\sqrt{x - 3} = 4$$

- The minus 3 is inside the radical, so do not add 3 first
- Square both sides to eliminate the radical symbol
- The equation becomes x minus 3 equals 16
- Add 3 to both sides to get x equals 19

Answer: $x = 19$



5. Solve for m.

$$\sqrt{3m+1} = \sqrt{2m+6}$$

- Both sides contain radicals, so square both sides to eliminate them
- The equation becomes 3m plus 1 equals 2m plus 6
- Subtract 2m from both sides to get m plus 1 equals 6
- Subtract 1 from both sides to get m equals 5

Answer: $m = 5$

6. Solve for x.

$$\sqrt{x+7} = 5$$

- The plus 7 is inside the radical, so square both sides first
- The equation becomes x plus 7 equals 25
- Subtract 7 from both sides
- The result is x equals 18

Answer: $x = 18$

7. Solve for x.

$$\sqrt{2x+1} = 3$$

- Square both sides to eliminate the radical
- The equation becomes 2x plus 1 equals 9
- Subtract 1 from both sides to get 2x equals 8
- Divide both sides by 2 to get x equals 4

Answer: $x = 4$

8. Solve for x.

$$\sqrt{x} - 6 = 2$$

- Add 6 to both sides to isolate the square root
- The equation becomes square root of x equals 8
- Square both sides to eliminate the radical
- Eight squared is 64, so x equals 64

Answer: $x = 64$

9. Solve for x.

$$\sqrt{4x-3} = 5$$

- Square both sides to eliminate the radical
- The equation becomes 4x minus 3 equals 25
- Add 3 to both sides to get 4x equals 28
- Divide both sides by 4 to get x equals 7

Answer: $x = 7$



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10. Solve for x.

$$\sqrt{x-1} + 2 = 6$$

- Subtract 2 from both sides to isolate the radical
- The equation becomes square root of x minus 1 equals 4
- Square both sides to eliminate the radical
- The equation becomes x minus 1 equals 16, so x equals 17

Answer: **$x = 17$**



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